VANISHING OF THE TOP LOCAL COHOMOLOGY OF ONE FINITE MODULE OF FULL SUPPORT FORCES THE COHOMOLOGICAL DIMENSION OF THE IDEAL TO DROP

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The vanishing of local cohomology above the dimension of the support and the bound cd(A, I) \leq dim(A). Author lemma lemma-cd-dimension, whose statement is reproduced verbatim as a context span of this card (source lines 758-764). Gives the a priori bound cd(A, I) \leq d and the vanishing that makes the property P descend along surjections.
- The statement that a property of finite modules which holds for a module of full support and is inherited by quotients of direct sums holds for every finite module. Invoked at the end of the counted proof (source lines 3626-3628) and attributed to a lemma of another chapter (a cross-chapter pointer). Spreads the vanishing from the given module to all finite modules, which is what cd(A, I) < d means.

Source reference. Original reference coherent-lemma-property-higher-rank-cohomological: sources/stacks-local-cohomology/pinned-original-source.tex, source lines 3627–3627. Navigation pointer only: the counted text refers to the Stacks item coherent-lemma-property-higher-rank-cohomological, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let A be a Noetherian ring of dimension d. Let $I \subset A$ be an ideal. If $H^d_{V(I)}(M) = 0$ for some finite A-module whose support contains all the irreducible components of dimension d, then $cd(A, I) < d$.*

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Lemma 0.2. *Let $I \subset A$ be a finitely generated ideal of a ring A. If M is a finite A-module, then $H^i_{V(I)}(M) = 0$ for $i > \dim(Supp(M))$. In particular, we have $cd(A, I) \leq \dim(A)$.*

Proof. By Lemma 0.2 we know $\mathrm{cd}(A, I) \leq d$. Thus for any finite A-module N we have $H^i_{V(I)}(N) = 0$ for $i > d$. Let us say property $\mathcal{P}$ holds for the finite A-module N if $H^d_{V(I)}(N) = 0$. One of our assumptions is that $\mathcal{P}(M)$ holds. Observe that $\mathcal{P}(N_1 \oplus N_2) \Leftrightarrow (\mathcal{P}(N_1) \wedge \mathcal{P}(N_2))$. Observe that if $N \to N'$ is surjective, then $\mathcal{P}(N) \Rightarrow \mathcal{P}(N')$ as we have the vanishing of $H^{d+1}_{V(I)}$ (see above). Let $\mathfrak{p}_1, \ldots, \mathfrak{p}_n$ be the minimal primes of A with $\dim(A/\mathfrak{p}_i) = d$. Observe that $\mathcal{P}(N)$ holds if the support of N is disjoint from $\{\mathfrak{p}_1, \ldots, \mathfrak{p}_n\}$ for dimension reasons, see Lemma 0.2. For each i set $M_i = M/\mathfrak{p}_i M$. This is a finite A-module annihilated by $\mathfrak{p}_i$ whose support is equal to $V(\mathfrak{p}_i)$ (here we use the assumption on the support of M). Finally, if $J \subset A$ is an ideal, then we have $\mathcal{P}(JM_i)$ as JM_i is a quotient of a direct sum of copies of M. Thus it follows from Cohomology of Schemes, Lemma coherent-lemma-property-higher-rank-cohomological that $\mathcal{P}$ holds for every finite A-module. $\square$